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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	**PnJ**
Project / Solution Name	**FutureReady**
Name (Leader)	**Jessy Chia Jiek Xin**
Email & Phone (Leader)	**jimhow00@gmail.com / 011-55596939**
University Affiliation(s)	**Multimedia University**
Live Prototype URL	**https://future-ready-pnj.replit.app**
Code Repository URL (public / access granted to judges)	**https://github.com/patricia5788/FutureReady_PnJ**

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

FutureReady is a web application developed and deployed on Replit. Its frontend uses React, TypeScript, Vite, and Tailwind CSS, while the backend uses Node.js and Express.

The platform enables users to assess their skills, explore career matches, identify skill gaps, and access personalized learning roadmaps. Career matching uses explainable, rule-based logic rather than AI-generated employment predictions.

User profiles and roadmap progress are stored in browser localStorage, while PostgreSQL supports asynchronous AI job tracking.

Optional AI-generated career insights are powered by the pre-trained Qwen3-4B-Instruct-2507 model through RunPod Serverless and vLLM. This feature is controlled by a server-side configuration flag and remains disabled by default in the public demo.

The application integrates the Remotive Public API to display sample remote job listings. Career and skills information is curated and adapted from O*NET OnLine and ESCO references, without live API integration. Google Fonts provides frontend typography.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript 5.9.3	Application programming	Apache-2.0
React 19.1.0	Frontend user interface	MIT

Name & Version	Purpose in Project	License
Vite 7.3.7	Frontend development and build	MIT
Express 5.2.1	Backend API server	MIT
Drizzle ORM 0.45.2	Database access	Apache-2.0
pg 8.23.0	PostgreSQL connection	MIT
Tailwind CSS 4.3.3	User interface styling	MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
RunPod Serverless API	Optional AI career insights	Sponsor-provided
Remotive Jobs API	Sample remote job listings	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Development environment and deployed web application hosting	Replit deployment; region not independently confirmed
RunPod Serverless	Optional GPU inference using Qwen	Serverless GPU inference; region not independently confirmed
PostgreSQL via Replit environment	Database for asynchronous AI job tracking	Database provider and region not independently verified
GitHub	Source code repository and version control	Public GitHub repository accessible to judges

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Qwen3-4B-Instruct-2507	Qwen	Pre-trained; inference only	Apache License 2.0

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Remote job listings	Remotive Public API	Public external data	Remotive API Terms; source credit and original links	N
Career and skill mapping data	Team-curated; informed by O*NET OnLine and ESCO	Team-curated / source-adapted	Team-created elements; applicable O*NET and ESCO terms	N
User-entered skills and proficiency	User input	Self-collected	Not applicable	Y
O*NET occupational data	O*NET OnLine	Public reference data	CC BY 4.0; attribution and modification notice	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
ESCO skills and occupations	European Commission ESCO	Public reference material	European Commission reuse conditions	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Replit Agent	Application development, debugging, integration and testing	Major
ChatGPT	Technical planning, troubleshooting and documentation assistance	Major

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Our team used AI-assisted development tools for implementation, debugging, technical planning, and documentation. We reviewed generated code and outputs, tested key application workflows, and verified TypeScript checks and production builds. We also tested career-matching behaviour, skill-gap updates, and the RunPod AI integration.

The team remains responsible for the accuracy, security, functionality, and final submission of the project. AI-generated suggestions were treated as assistance rather than independently verified facts.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Platform / development credits	Application development, testing and public deployment
RunPod	GPU compute credits / Serverless platform	Hosting and testing Qwen AI inference

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit supports the application's development, testing, and hosting workflow, while RunPod provides GPU inference for optional AI-generated career insights.

The core career-matching, skill-gap analysis, learning roadmap, and SDG mapping features use rule-based logic and can continue operating without RunPod AI inference.

The AI inference provider could be replaced by another compatible service, subject to technical integration and cost considerations. Replit hosting could also be replaced by another compatible deployment platform.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

User skill profiles and learning-roadmap progress are stored locally in the user's browser. When optional AI-generated career insights are requested, selected career information, skill names and levels, calculated skill gaps, and limited job-listing summaries may be transmitted to the configured RunPod inference endpoint.

The application does not intentionally send users' education details, personal goals, interests, or contact information to RunPod as part of this feature.

PostgreSQL stores AI job-tracking information, generated responses, statuses, and timestamps. AI job records have a 24-hour logical expiry, but automatic physical deletion of all expired records has not been implemented.

API credentials are accessed through environment variables rather than embedded directly in application code. Provider-side data retention practices require separate verification.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

AI-generated career insights may be inaccurate or biased. Career-match scores are rule-based alignment indicators, not hiring probabilities.

Remotive job listings and AI-adoption scenarios are illustrative rather than verified labour-market forecasts.

RunPod inference may experience cold starts or service interruptions and is disabled by default in the public demo.

AI job records have a 24-hour logical expiry, but automatic physical deletion is not implemented.

The current dependency audit reports a high-severity Zod advisory and a moderate build-tooling advisory; remediation remains pending.

The prototype does not guarantee employment outcomes.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Jessy Chia

25/10/26